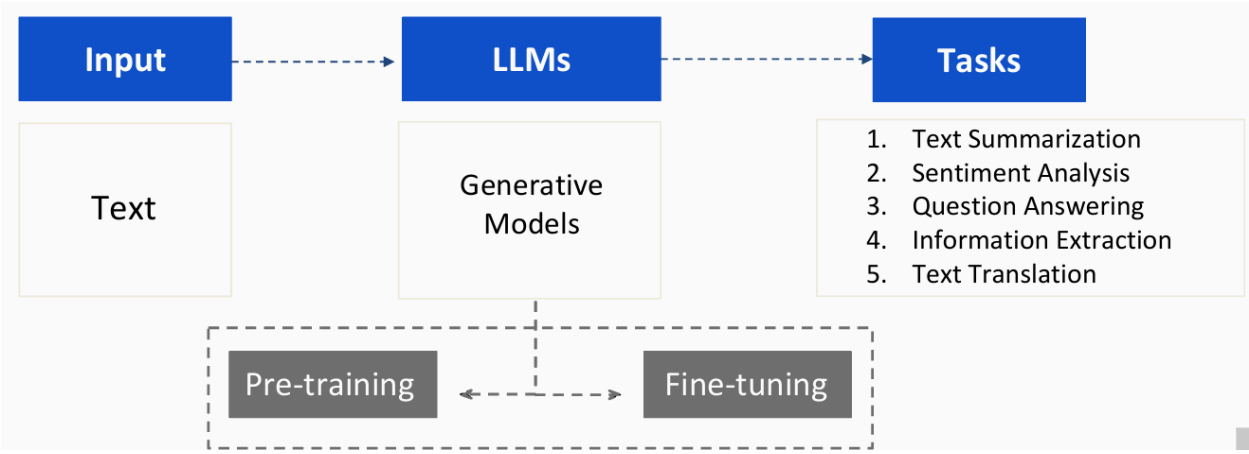
Understanding LLMs: How They Work

What is an LLM?

LLM stands for Large Language Model. It's a type of artificial intelligence (AI) designed to understand and generate human-like text. Think of it as a super-smart computer program that can read, write, and even chat with you in a way that feels natural.

How does an LLM work?

1. **Training on Data:**
    - LLMs are trained on huge amounts of text data (like books, articles, websites, etc.).
    - They learn patterns, grammar, facts, and even some reasoning skills from this data.
2. **Predicting Words:**
    - When you ask an LLM a question or give it a prompt, it predicts what words should come next based on what it has learned.
    - For example, if you type "The sky is...", it might predict "blue" because it has seen that phrase many times.
3. **Generating Responses:**
    - It doesn't just repeat what it has seen. Instead, it combines what it knows to create new, coherent sentences that make sense in the context.
4. **Fine-Tuning:**
    - After the initial training, LLMs can be fine-tuned for specific tasks, like answering questions, writing essays, or even coding.



In Simple Terms:

An LLM is like a very advanced autocomplete tool. It uses what it has learned from reading tons of text to help you with writing, answering questions, or having a conversation. It doesn't "think" like a human, but it's really good at mimicking how we use language.

Example:

If you ask, "What is the capital of France?", the LLM will predict the answer is "Paris" because it has seen that information many times in its training data.

1. Setting Up ChatGPT (by OpenAI)

ChatGPT is accessible via OpenAI's platform or API. Here's how to set it up:

1. Create an OpenAI Account:
    - Go to https://chatgpt.com/
    - Sign up for an account using your email or Google/Microsoft account.
2. Access ChatGPT:
    - Once logged in, navigate to https://chatgpt.com/
    - You can start using ChatGPT for free (with limitations) or subscribe to ChatGPT Plus for advanced features.
3. Start Chatting:

- Type your questions or prompts in the chatbox, and ChatGPT will respond.

2. Setting Up Google Gemini

Google Gemini is part of Google's AI ecosystem. As of now, Gemini is not publicly available, but you can use Google's AI tools like Bard or Vertex AI for similar functionality.

1. Access Google Bard:
   - Go to https://gemini.google.com/
   - Sign in with your Google account.
2. Start Chatting:
   - Type your prompts, and Gemini will respond.

3. Setting Up DeepSeek

DeepSeek is a Chinese AI platform. Here's how to set it up:

1. Visit DeepSeek:
   - Go to https://www.deepseek.com/
   - Sign up for an account.
2. Access the Chat Interface:
   - Use the pre-built chat interface to interact with DeepSeek.

Sample Prompts used:

Prompt 1: write test cases for a shopping cart feature

Prompt 2: Generate test cases for an ATM machine

Prompt 3: Write a test automation script in selenium java for the following steps. 1) Open google.com 2) search for text selenium 3) click on search button 4) validate results.